Improving the accuracy of simulation of radiation-reaction effects with implicit Runge-Kutta-Nyström methods

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The Landau-Lifshitz equation provides an efficient way to account for the effects of radiation reaction without acquiring the nonphysical solutions typical for the Lorentz-Abraham-Dirac equation. We solve the Landau-Lifshitz equation in its covariant four-vector form in order to control both the energy and momentum of radiating particles. Our study reveals that implicit time-symmetric collocation methods of the Runge-Kutta-Nyström type are superior in accuracy and better at maintaining the mass-shell condition than their explicit counterparts. We carry out an extensive study of numerical accuracy by comparing the analytical and numerical solutions of the Landau-Lifshitz equation. Finally, we present the results of the simulation of particle scattering by a focused laser pulse. Due to radiation reaction, particles are less capable of penetrating into the focal region compared to the case where radiation reaction is neglected. Our results are important for designing forthcoming experiments with high intensity laser fields.

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I. INTRODUCTION

In recent years the problem of radiation reaction in the realm of high intensity laser fields has attracted much interest. A rigorous approach to radiation reaction requires quantum treatment of the stochastic events of hard photon emission [1–4]. In numerical simulation this was incorporated by using Monte Carlo sampling of photon emission [5]. This method treats correctly the recoil due to emission of hard photons with $\hbar\omega > mc^2 \sim 0.511$ MeV (m is the electron mass, and c is the speed of light) but is computationally rather expensive. At the same time, the relatively soft part of the photon spectrum, 10 keV to 0.511 MeV, had to be ignored in plasma simulations due to a limited numerical resolution.

One of the approaches to account for the radiation effect due to soft photon emission is to consider it classically by adding the radiation friction term into the equation of particle motion. The classical problem of motion of a radiating charge was considered in [6], where the so-called Lorentz-Abraham-Dirac (LAD) equation was derived. This equation of motion contains third order derivative $\ddot{x}^\mu$ and therefore possesses problematic runaway or causality violating solutions. To ensure physical behavior the LAD equation requires careful choice of both initial and boundary conditions (see [7,8] for a review), which is difficult to realize in computational practice. In the most of physically meaningful situations, radiation reaction can be treated as a perturbation. The first order of perturbation theory results in the Landau-Lifshitz (LL) equation [9,10]. This equation contains nonlinear dissipation, which leads to contraction of the phase space volume [11]. The impact of radiation reaction in the classical regime attracts attention in various fields of modern physics ranging from high energy astrophysics [12] to such hot topics of laser physics as electron dynamics [11,13] and radiation [14,15] in a strong laser field, cascade production [5], electron [16] and ion [17,18] acceleration, etc. A brief summary of typical numerical implementation was given in Ref. [14]. A more comprehensive review and references can be found in [10,19].

In plasma simulation solvers [20] the equations of particle motion are usually integrated by the modified Störmer-Verlet method [21] known also as the leapfrog scheme. This originally semi-implicit method can be turned into an explicit one provided that the special ansatz [20,22] is used to treat properly the rotation in a magnetic field. However, inclusion of the extra term responsible for the radiation reaction breaks the symmetry of the relativistic Newton-Lorentz (NL) equation. Since radiation reaction depends upon momenta nonlinearly, the explicitness of the method can no longer be maintained within the second order of accuracy. On the other hand, both accuracy and explicitness can be restored to some extent if the predictor-corrector technique is applied for the dissipative Landau-Lifshitz equation. Such an approach was undertaken in [17,18] to introduce the radiation friction force into the particles-in-cell simulation of the ion acceleration.

Another concern with simulation of the covariant LL equation is maintaining the mass shell condition (conservation of the Minkowski norm $u_\mu u^\mu = c^2$). The Minkowski norm can be preserved exactly in the course of numerical simulation if the corresponding equation of motion is lifted onto the Lie group settings [23,24]. The Lie group methods show promising results but are rather expensive for nonlinear equations, as they require multiple computations of the matrix exponentials for each time step [25].

In this paper we would like to draw attention to the Runge-Kutta(-Nyström) [RK(N)] type methods [26], which are specially designed for solving the second order ordinary differential equations (ODEs). However, the explicit RK(N)-type methods do not conserve exactly the quadratic integrals of motion (e.g., energy and the Minkowski norm) and hence are not suitable enough for integration over a long time. Contrarily, the implicit collocation RK(N) methods [27–29] provide better accuracy and are capable of conserving quadratic integrals of motion. This, together with time reversibility, singles out

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these methods as perfect candidates for integration of the LL equation over a long time.

This paper is organized as follows. In Sec. II a summary of covariant formulation of the equation of motion of a radiating particle is presented. The main goal of the paper is achieved in Sec. III, where we present the derivation of the collocation Gauss-Legendre RKN method entirely in terms of collocation points for the fourth, sixth, and eighth orders of accuracy. The result is employed in Secs. IV and V, where we perform extensive validation studies of the considered RK(N) methods against exact solutions of the LL equation in a constant magnetic field and in a plane wave and demonstrate that implicit collocation methods of RK(N) type are superior in accuracy to their explicit counterparts in solving both the NL and LL equations. In Sec. VI we present an important application of the developed numerical technique to the problem of relativistic scattering of electrons by a focused laser field. Our conclusions are summarized in Sec. VII.

II. BASIC EQUATIONS

Radiation reaction modifies the NL equation by a radiation friction term g^μ on the right-hand side,

$$\frac{d^2x^\mu}{d\tau^2} = f^\mu = \frac{e}{m}F^\mu_\nu u^\nu + g^\mu. \tag{1}$$

The four-velocity $u^\mu = \{\gamma, \mathbf{u}\}$ of a charge is the derivative of the position four-vector $\mathbf{x}^\mu = \{t, \mathbf{r}\}$ with respect to the proper time τ. The electromagnetic field tensor F^μ_ν is expressed by the following matrix:

$$F^\mu_\nu = \begin{bmatrix} 0 & E_x & E_y & E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}$$

Radiation reaction was originally considered with the LAD equation [6],

$$\frac{d^2x^\mu}{d\tau^2} = \frac{e}{m}F^\mu_\nu u^\nu + \frac{2}{3}\frac{e^2}{m}\left(u_\nu \frac{d^2u^\mu}{d\tau^2} - u^\mu \frac{d^2u_\nu}{d\tau^2}\right)u^\nu. \tag{2}$$

Here and in the following we adopt the units $c = \hbar = 1$, $e = \sqrt{\alpha}$, where $\alpha = e^2/\hbar c \simeq 1/137$ is the fine structure constant. As pointed out in the Introduction, Eq. (2) possesses the problematic runaway or causality violating solutions. The regular way to avoid such an unwanted solution is to treat the second term on the right-hand side as a perturbation [9]. Then by virtue of

$$\dot{F}^\mu_\nu = \frac{dx^\lambda}{d\tau}\frac{\partial F^\mu_\nu}{\partial x^\lambda} = u^\lambda \partial_\lambda F^\mu_\nu,$$

the term g^μ takes the form

$$g^\mu = \frac{2e^3}{3m^2}u^\lambda\left(\partial_\lambda F^\mu_\nu\right)u^\nu + \frac{2e^4}{3m^3}\left[F^\mu_\nu F^\nu_\lambda u^\lambda + (F_{kl}u^l)\left(F^k_m u^m\right)u^\mu\right]. \tag{3}$$

Equation (1) with the term g^μ given by (3) is known as the LL equation. The numerical solution of the LL equation should preserve the Minkowski norm $u^\mu u_\mu = 1$ as well as the orthogonality condition $g^\mu u_\mu = 0$. It turns out that actual fulfillment of the former condition strongly depends on a chosen numerical method, while implementation of the latter one is difficult to enforce if g^μ is in the form (3). Indeed, in this form g^μ consists of three terms, which contribute disparately. While in the majority of situations the first term in (3) can be safely neglected, the two remaining terms together do satisfy the orthogonality condition, but due to a huge difference in the magnitude of their components naive summation would lead to accumulation of considerable round-off errors. This purely numerical problem may cause eventual violation of orthogonality between the four-force and four-momentum. In order to avoid such an unwanted effect we consider the LL equation in the "symmetrized" form reminiscent of the LAD equation. So the actual equation which is solved in this paper is given as follows:

$$\frac{d^2x^\mu}{d\tau^2} = \frac{e}{m}F^\mu_\nu u^\nu + \frac{2}{3}\frac{e^2}{m}(u_\nu \dot{w}^\mu - u^\mu \dot{w}_\nu)u^\nu, \tag{4}$$

where $\dot{w}^\mu$ is calculated as

$$\dot{w}^\mu = \frac{e}{m}\left(\dot{F}^\mu_\nu u^\nu + F^\mu_\nu \dot{u}^\nu\right) = \frac{e}{m}\left[u^\lambda\left(\partial_\lambda F^\mu_\nu\right)u^\nu + \frac{e}{m}F^\mu_\nu F^\nu_\lambda u^\lambda\right].$$

In Eq. (4) both leading radiation reaction terms acquire the same order, so that their subtraction can no longer cause accumulation of round-off errors. Such a trick allows us to keep $g^\mu u_\mu = 0$ satisfied up to the level of machine precision.

III. NUMERICAL INTEGRATION OF THE MOTION OF A CHARGE

A. Explicit Runge-Kutta-Nyström methods

The most frequently preferred approach to solve a second order ODE $\ddot{x}^\mu = f^\mu(\tau, x^\mu, \dot{x}^\mu)$ with higher accuracy is to apply a suitable method to an equivalent system of the first order system of ODEs,

$$\frac{d}{d\tau}\begin{pmatrix} x^\mu \\ u^\mu \end{pmatrix} = \begin{pmatrix} u^\mu \\ f^\mu(\tau,\, x^\mu,\, u^\mu) \end{pmatrix}. \tag{5}$$

Application of a RK method to such a problem results in [30]

$$L^\mu_i = u^\mu_n + h\sum_{j=1}^{s} A_{ij}K^\mu_j, \quad i = 1, \ldots, s, \tag{6}$$

$$K^\mu_i = f^\mu\left(\tau_n + hc_i,\, x^\mu_n + h\sum_{j=0}^{s} A_{ij}L_j,\, u^\mu_n + h\sum_{j=1}^{s} A_{ij}K_j\right), \quad i = 1, \ldots, s, \tag{7}$$

$$x^\mu_{n+1} = x^\mu_n + h\sum_{j=1}^{s} a_j L^\mu_j, \tag{8}$$

$$u^\mu_{n+1} = u^\mu_n + h\sum_{j=1}^{s} a_j K^\mu_j, \tag{9}$$

TABLE I. Butcher tables for the (a) RK and (b) RKN methods.

$\boldsymbol{c}$	$A_{s\times s}$
	$\boldsymbol{a}^T$

(a) RK

$\boldsymbol{c}$	$\mathbf{A}_{s\times s}$	$\mathbf{B}_{s\times s}$
	$\boldsymbol{a}^T$	$\boldsymbol{b}^T$

(b) RKN

where h is a step size and A_{ij}, c_i, and a_i are the scheme-specific coefficients [see Table I(a)].

An alternative approach to solve the equation of motion is to explore the original second order ODE $\ddot{x}^\mu = f^\mu(\tau,x^\mu,\dot{x}^\mu)$ without first reducing it to a system of the first order ODEs. This can be achieved by eliminating L_i from formulas (6)–(9) as follows:

$$K_i^\mu = f^\mu\left(\tau_n + c_i h, x_n^\mu + c_i h u_n^\mu + h^2\sum_{j=1}^{s} B_{ij}K_j^\mu, u_n^\mu + \sum_{j=1}^{s} A_{ij}K_j^\mu\right), \tag{10}$$

$$x_{n+1}^\mu = x_n^\mu + h u_n^\mu + h^2\sum_{j=1}^{s} b_j K_j^\mu, \tag{11}$$

$$u_{n+1}^\mu = u_n^\mu + h\sum_{j=1}^{s} a_j K_j^\mu, \tag{12}$$

where the coefficients B_{ij} and b_j are given by

$$B_{ij} = \sum_k A_{ik}A_{kj}, \quad b_j = \sum_k a_k A_{kj}. \tag{13}$$

The conditions (13) for the new coefficients B_{ij}, b_j in fact can be abandoned, as was found by Nyström [26]. This kind of method is known as the RKN method (see also [31]). In the most general form the RKN methods are defined by Eqs. (10)–(12), where the coefficients A_{ij}, B_{ij}, a_i, b_i are kept independent and form the double Butcher array [see Table I(b)]. If $A_{ij} = 0$ and $B_{ij} = 0$ for $j \geqslant i$, then such a construction yields an explicit method, in which each stage K_i^μ can be determined sequentially. The examples of explicit fourth order RK and RKN methods are given in Table II.

B. Implicit collocation Runge-Kutta-Nyström methods

The basic idea of the collocation methods [27] is to select polynomials $r^\mu(\tau)$ of degree $s+1$ that satisfy the initial value problem at $s+1$ points $\tau_n, \tau_n + c_1 h, \ldots, \tau_n + c_s h \in [\tau_n, \tau_{n+1})$, where $c_1, \ldots, c_s$ are the distinct real numbers from the interval (0, 1). Then for the second order ODE $\ddot{x}^\mu = f^\mu(\tau,x^\mu,\dot{x}^\mu)$ the corresponding polynomial approximation r^μ satisfies [29]

$$\begin{aligned} r^\mu(\tau_n) = x_n^\mu, \quad \dot{r}^\mu(\tau_n) = u_n^\mu, \\ \ddot{r}^\mu(\tau_n + c_i h) = f^\mu(\tau_n + c_i h,\, r^\mu(\tau_n + c_i h), \\ \dot{r}^\mu(\tau_n + c_i h)) \quad (i = 1, \ldots, s). \end{aligned} \tag{14}$$

In the following we proceed with a derivation of an RKN method (the corresponding RK method is defined by a subtable with the coefficients A_{ij}). Being of degree $s-1$, the second derivative of the collocation polynomial $\ddot{r}^\mu(\tau_n + \xi h)$ can be uniquely expanded in the Lagrange polynomials

$$L_i(\xi) = \prod_{k\neq i}^{s} \frac{\xi - c_k}{c_i - c_k}. \tag{15}$$

Setting $K_i^\mu = \ddot{r}^\mu(\tau_n + c_i h)$, this expansion can be written as

$$\ddot{r}^\mu(\tau_n + \xi h) = \sum_{j=1}^{s} L_j(\xi) K_j^\mu, \quad \tau_n + \xi h \in [\tau_n,\, \tau_n + h].$$

This implies that $\dot{r}^\mu$ and r^μ can be approximated on $[\tau_n,\, \tau_n + h]$ by

$$\dot{r}^\mu(\tau_n + \xi h) = u_n^\mu + \sum_{i=1}^{s} K_i^\mu A_i(\xi), \quad A_i(\xi) = \int_0^\xi L_j(\xi')d\xi', \tag{16}$$

$$r^\mu(\tau_n + \xi h) = x_n^\mu + \xi h u_n^\mu + \sum_{i=1}^{s} K_i^\mu B_i(\xi),$$

$$B_i(\xi) = \int_0^\xi A_i(\xi')d\xi' = \int_0^\xi (\xi - \xi')L_i(\xi')d\xi'. \tag{17}$$

Now, by substituting $\xi = c_i$ for successive $i = 1, \ldots, s$, we can make formulas (14), (16), and (17) coincide with Eqs. (10)–(12) and thus specify the discrete coefficients of the collocation method:

$$A_{ij} = A_j(c_i), \quad a_j = A_j(1), \quad B_{ij} = B_j(c_i), \quad b_j = B_j(1). \tag{18}$$

The fourth order method can be obtained by using two Lagrange polynomials of degree 1:

$$L_1(\xi) = \frac{\xi - c_2}{c_1 - c_2}, \quad L_2(\xi) = \frac{\xi - c_1}{c_2 - c_1}. \tag{19}$$

The resulting method is presented in Table III(a). The highest possible order $2s$ can be achieved when the collocation points are chosen as roots of the sth shifted Legendre polynomial [28],

$$\frac{d^s}{d\tau^s}[\tau^s(\tau - 1)^s] = 0, \quad \tau \in (0,\, 1). \tag{20}$$

Other possible choices of the collocation points (Gauss-Lobatto and Gauss-Radau methods) possess lower order $p < 2s$ but might still be beneficial if, e.g., the equations are stiff [32]. For $s = 2$ the roots of the quadratic polynomial (20)

TABLE II. Explicit (a) RK and (b) RKN methods of order 4.

0	0	0	0	0
1/2	1/2	0	0	0
1/2	0	1/2	0	0
1	0	0	1	0
	1/6	1/3	1/3	1/6

(a) RK

0	0	0	0	0	0	0	0	0	0	0	0	0
1/2	1/2	0	0	0	1/2	1/2	0	0	0	1/8	0	0	0
1/2	0	1/2	0	0	1/2	0	1/2	0	0	1/8	0	0	0
1	0	0	1	0	1	0	0	1	0	0	0	1/2	0
	1/6	1/3	1/3	1/6		1/6	1/3	1/3	1/6	1/6	1/6	1/6	0

(b) RKN

TABLE III. Double Butcher table for (a) the collocation and (b) Gauss-Legendre RKN methods for $s = 2$.

c_1	$\frac{c_1(c_1-2c_2)}{2(c_1-c_2)}$	$\frac{c_1^2}{2(c_1-c_2)}$	$\frac{c_1^2(c_1-3c_2)}{6(c_1-c_2)}$	$\frac{c_1^3}{3(c_1-c_2)}$
c_2	$\frac{c_2^2}{2(c_2-c_1)}$	$\frac{c_2(c_2-2c_1)}{2(c_2-c_1)}$	$\frac{c_2^3}{3(c_2-c_1)}$	$\frac{c_2^2(c_2-3c_1)}{6(c_2-c_1)}$
	$\frac{1-2c_2}{2(c_1-c_2)}$	$\frac{1-2c_1}{2(c_2-c_1)}$	$\frac{1-3c_2}{6(c_1-c_2)}$	$\frac{1-3c_1}{6(c_2-c_1)}$

(a) Collocation RKN

$\frac{1}{2}-\frac{\sqrt{3}}{6}$	$\frac{1}{4}$	$\frac{1}{4}-\frac{\sqrt{3}}{6}$	$\frac{1}{36}$	$\frac{5}{36}-\frac{\sqrt{3}}{12}$
$\frac{1}{2}+\frac{\sqrt{3}}{6}$	$\frac{1}{4}+\frac{\sqrt{3}}{6}$	$\frac{1}{4}$	$\frac{5}{36}+\frac{\sqrt{3}}{12}$	$\frac{1}{36}$
	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}+\frac{\sqrt{3}}{12}$	$\frac{1}{4}-\frac{\sqrt{3}}{12}$

(b) Gauss-Legendre RKN

are $c_{1,2} = 1/2 \mp \sqrt{3}/6$, and the resulting Gauss-Legendre RKN method of order 4 is presented in Table III(b).

In this paper we also use higher order methods based on $s = 3$ and $s = 4$ collocation points. The Gauss-Legendre RKN method with $s = 3$ collocation points is briefly described in the Appendix. Although computationally more expensive, these sixth and eighth order methods provide the possibility to measure the accuracy of lower order methods in case an analytical solution is not available.

After the coefficients (18) are substituted into the system (10)–(12) it turns into a set of stage equations. The implicitness of the collocation RK(N) methods implies usage of the iteration technique to solve the stage equations. The simplest iteration scheme can be obtained by just hanging the iteration indices $m, m+1$ on the implicit equations:

$$\begin{aligned} \text{iRK(N):}\quad & K_i^{\mu,m+1} \\ &= f^\mu\left(\tau_n + c_i h,\, x_n^\mu + c_i h u_n^\mu \right. \\ &\left. + h^2 \sum_{j=1}^{s} B_{ij} K_j^{\mu,m},\, u_n^\mu + h \sum_{j=1}^{s} A_{ij} K_j^{\mu,m}\right), \end{aligned} \tag{21}$$

where the prefix "i" stands for implicit to distinguish the method from explicit ones, for which we will similarly use the prefix "e" in front of the abbreviation RK(N). In order to formulate the criterion for stopping the iteration process, an error of the approximation has to be measured by an appropriate norm. In this paper we use the maximum norm for this purpose,

$$||K^{m+1} - K^m||_\infty = \max_{i,\mu} \left|K_i^{\mu,m+1} - K_i^{\mu,m}\right|. \tag{22}$$

In the rest of the paper, we discuss applications of the described numerical methods to physical problems of particle motion in ultrastrong electromagnetic fields.

IV. PARTICLE IN MAGNETIC FIELD

Let us start with the motion of an ultrarelativistic particle in a constant magnetic field. By fixing the direction of magnetic field along $x^3 = z$, the LL equation reads

$$\frac{d\boldsymbol{u}_\perp}{d\tau} = \frac{e}{m}\boldsymbol{u}_\perp \times \boldsymbol{H} - \frac{2}{3}\frac{e^4}{m^3} H^2 (1+u_\perp^2)\boldsymbol{u}_\perp, \tag{23}$$

$$\frac{du_z}{d\tau} = -\frac{2}{3}\frac{e^4}{m^3} H^2 u_\perp^2 u_z, \tag{24}$$

where $\boldsymbol{u}_\perp = \{u_x, u_y\}$ is the transverse component of four-velocity. After changing to polar coordinates $u_x = u_\perp \cos\phi$, $u_y = u_\perp \sin\phi$ in the transverse plane xy and introducing dimensionless time $\varphi = \Omega\tau$ (where $\Omega = eH/m$ is the rotation frequency in the proper reference frame), the system (23) and (24) takes the form

$$\frac{du_\perp}{d\varphi} = -K u_\perp (1+u_\perp^2), \tag{25}$$

$$\frac{d\phi}{d\varphi} = -\sigma, \tag{26}$$

$$\frac{du_z}{d\varphi} = -K u_\perp^2 u_z. \tag{27}$$

Here $\sigma = \pm 1$ is the sign of a particle charge, and K is the dimensionless parameter

$$K = \frac{2}{3}\frac{e^2\Omega}{m} = \frac{2}{3}\frac{e^3 H}{m^2} = \frac{2}{3}\frac{\alpha H}{H_c}, \tag{28}$$

where $H_c = m^2/e$ is the critical magnetic field. Note that under the applicability of the LL equation $K \ll 1$. Equation (25) admits separation of variables; after that Eqs. (26) and (27) can also be easily solved. The solution reads

$$\begin{aligned} \boldsymbol{u}(\varphi) = & \frac{1}{\sqrt{(1+u_{\perp 0}^2)\, e^{2K\varphi} - u_{\perp 0}^2}} \\ & \times \{u_{\perp 0}\cos(\phi_0 - \sigma\varphi),\, u_{\perp 0}\sin(\phi_0 - \sigma\varphi),\, u_{z0} e^{K\varphi}\}, \end{aligned} \tag{29}$$

where the subscript 0 refers to the initial values at $\varphi = 0$. By simple integration one can find the particle trajectory in terms of the hypergeometric function; for example, its projection onto the plane xy is given by

$$\begin{aligned} & \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} \\ & = \frac{\sigma}{u_{\perp 0}} \left\{ \sqrt{(1+u_{\perp 0}^2)\, e^{2K\varphi} - u_{\perp 0}^2} \begin{bmatrix} \mathrm{Im} \\ -\mathrm{Re} \end{bmatrix} e^{i(\phi_0 - \sigma\varphi)}\, {}_2F_1 \right. \\ & \left. \times \left(1,\, \frac{1}{2} - \frac{i\sigma}{2K},\, 1 - \frac{i\sigma}{2K},\, \frac{(1+u_{\perp 0}^2)}{u_{\perp 0}^2} e^{2K\varphi}\right) \right\}\Bigg|_0^\varphi \end{aligned} \tag{30}$$

(below we assume $u_{z0} = 0$). The trajectory is a converging spiral shown in Fig. 1 (left panel). The trajectory without radiation reaction corresponds to the case $K = 0$ and represents a circle. Numerical integration is performed with the RK method since this particular problem does not contain dependence on coordinates. The observed order of accuracy of the method can

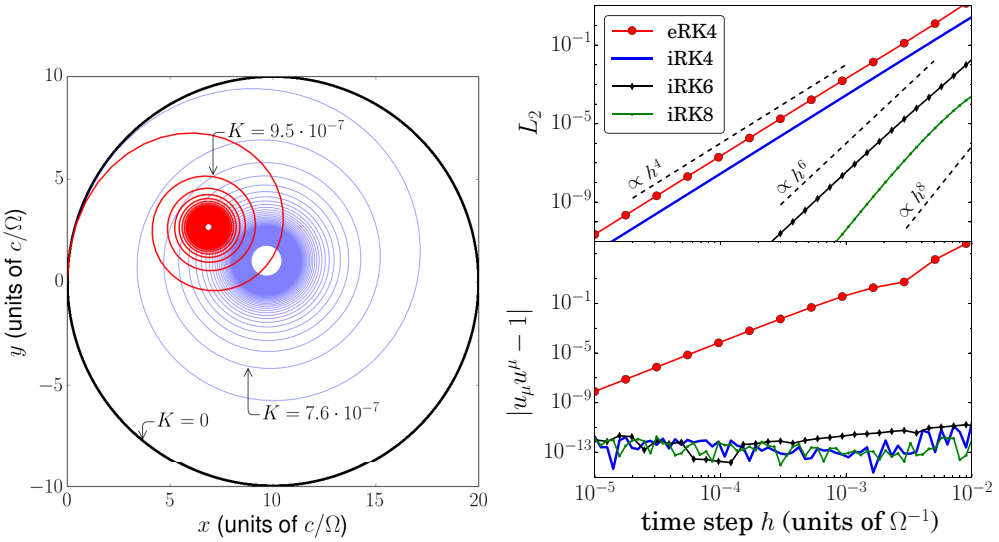


FIG. 1. (Color online) Effect of the radiation reaction on particle motion in a constant magnetic field with the initial Lorentz factor $\gamma_0 = 10^3$. (left) The particle trajectories for different values of the parameter $K = (2/3)\alpha H/H_c$. (right) The corresponding dependence of the error measure L_2 (top) and inaccuracy in preservation of the on-shell condition (bottom) on a time step.

be estimated by examining the behavior of the L_2 norm,

$$L_2 = \sqrt{\frac{h}{T}\sum_i \left|u^\mu(\tau_i) - u^\mu_{\rm ex}(\tau_i)\right|^2}, \tag{31}$$

where T is the proper time duration of the simulation and $u^\mu_{\rm ex}$ refers to the exact solution (29).

The results of studies of convergence of different numerical methods are presented in Fig. 1 (right panel), where both L_2 and the Minkowski norm errors are plotted depending upon a time step h. The slopes of the curves $L_2(h)$ in the right panel of Fig. 1, corresponding to both the explicit eRK4 and implicit iRK4 methods, clearly correspond to order 4, as expected, but the difference in the initial error levels makes the implicit method more favorable with respect to higher accuracy. The higher order methods iRK6 and iRK8 also exhibit the expected slopes, corresponding to orders 6 and 8, respectively. The inaccuracy of preservation of the mass shell condition, as shown in the bottom right panel of Fig. 1, remains low and bounded for all implicit methods, while the eRK method results in substantial violation of the Minkowski norm. It should be noted that nonconservation of the Minkowski norm depends on the numerical error δu^μ as

$$(u^\mu + \delta u^\mu)(u_\mu + \delta u_\mu) = 1 + 2u^\mu \delta u_\mu + O(\delta u^2)$$

and hence is especially pronounced in the ultrarelativistic case $u^\mu \gg 1$.

V. PARTICLE IN PLANE WAVE FIELD

The LL equation admits a much simpler form for a particle moving in a plane wave field $A_\mu(x) = A_\mu(\varphi)$, $\varphi = kx = \omega_0 t - \boldsymbol{k}\cdot\boldsymbol{r}$, $kA = 0$:

$$\frac{du^\mu}{d\varphi} = \frac{e}{m}[h(A'u)k^\mu - A^{\mu\prime}] + \frac{2}{3}\frac{e^3}{m^2 h}[h(A''u)k^\mu - A^{\mu\prime\prime}] - \frac{2}{3}\frac{e^4}{m^3}A'^2 k^\mu + \frac{2}{3}\frac{e^4}{m^3 h}A'^2 u^\mu, \tag{32}$$

where $1/h = \dot\varphi = k_\mu u^\mu$. This equation can be solved analytically in quadratures [33], and in special cases the four-velocity u^μ can be even expressed in terms of elementary functions [34]. Here we consider two important cases corresponding to linear and circular polarization of the wave. The plane wave is taken in the form

$$\begin{aligned} \boldsymbol{E} &= \{0,\ -a_0\cos(\varphi),\ \varkappa a_0 \sin(\varphi)\},\\ \boldsymbol{B} &= \{0,\ -\varkappa a_0 \sin(\varphi),\ a_0\cos(\varphi)\}, \end{aligned} \tag{33}$$

where $\varkappa = 0$ ($\varkappa = 1$) corresponds to linear (circular) polarization. The effect of radiation reaction on a particle trajectory is most transparent in a (denoted by a tilde) reference frame comoving with the guiding center. In this frame the radiation-free trajectories would take an especially simple form: in the case of linear polarization, particles move along a figure eight, whereas in the case of circular polarization, the trajectory is just a circle. The velocity of a tilde frame is directed along the x axis and equals $V_{\rm l} = a_0^2/(4 + a_0^2)$ and $V_{\rm c} = a_0^2/(1 + a_0^2)$ for $\varkappa = 0$ and $\varkappa = 1$, respectively. The radiation reaction effects open a closed figure eight or circular motion and lead to a drift along the wave vector of the wave (see Fig. 2).

In order to gain more knowledge about the accuracy of the numerical methods we compare numerical and analytical solutions in the case of a circularly polarized plane wave, for which the dependence of the phase φ and therefore of u^μ and x^μ on the proper time τ can be expressed explicitly by [34]

$$\phi(\tau) = \frac{\sqrt{1 + 2a_0^2(ku_0)^2 K\tau/\omega_0} - 1}{a_0^2(ku_0)K/\omega_0}, \tag{34}$$

where K is this time defined by $K = (2/3)e^2\omega_0/m$. The results are presented in Fig. 3. One can see that the slopes of the curves in Fig. 3 (left panel) are in agreement with the expected orders of the methods, indicated by dashed lines. Only the curve for the eRKN4 method on this plot deviates slightly from power law when the time step becomes larger. As for the right panel of Fig. 2, it demonstrates the same behavior of the methods

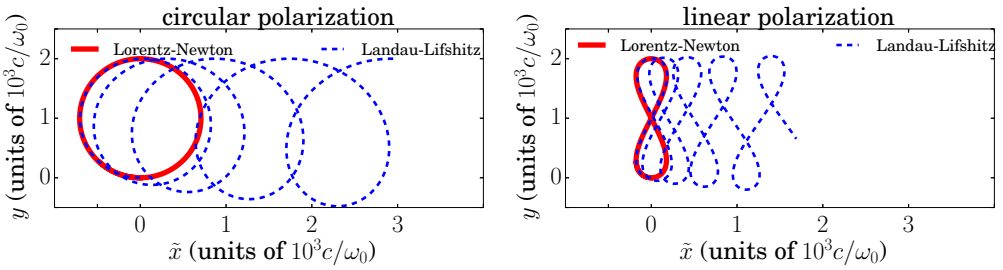


FIG. 2. (Color online) Projections of trajectories of a charge in a (left) circularly and (right) linearly polarized plane wave field for $a_0 = 10^3$ in the reference frame comoving with a guiding center in the absence of radiation reaction. Radiation reaction leads to a drift of the guiding center along the wave vector.

previously pointed out for the problem of motion in a constant magnetic field (compare to the right panel of Fig. 1).

Let us discuss computational complexity of the implicit methods. The main source of computational complexity in implicit methods comes from the number of iterations needed for a nonlinear solver to achieve the desired accuracy. Hence we measure it for both the RK and RKN methods using an average number of iterations per step. The results for the case of motion in the field of a linearly polarized plane wave are summarized in Fig. 4.

According to Fig. 4, the iRKN method requires fewer iterations than iRK. One can also compare qualitatively the performance of implicit iRKN4 and its explicit counterpart eRKN4. The number of stage equations (10) for eRKN4 equals 4, while the corresponding number of stages for iRKN4 is 2. But because there are four iterations on average for the same time step, the iRKN4 method is two times more expensive in the number of calls of the right-hand-side function. On the other hand, the implicit methods admit longer time steps without deterioration of numerical accuracy. Moreover, a further increase of the order of explicit methods eventually results in reaching the so-called Butcher barrier [28], after which the number of stages starts to grow faster than the order.

VI. SCATTERING OF PARTICLES ON FOCUSED LASER PULSES

In this section we apply the developed numerical technique to the important problem of scattering of particles by focused laser pulses. In the absence of radiation reaction, this problem can be studied with the theory of ponderomotive scattering (see, e.g., [35,36]). In previous studies of this problem radiation reaction was neglected because of the assumption that both the intensity and the particle energy were relatively small. Our basic settings are similar to those considered in [36], but with the laser intensity and particle energy two orders of magnitude higher, $\gamma_0, a_0 \sim 10^3$. Consider first a focused monochromatic Gaussian beam with the electric field defined by

$$\boldsymbol{E} = \frac{iE_0 b^2}{(b + i\boldsymbol{r}\cdot\boldsymbol{n})^2}\left\{\boldsymbol{\varepsilon} + \frac{\omega[\boldsymbol{r}\cdot(\boldsymbol{\varepsilon}\times\boldsymbol{n})](\boldsymbol{r}\times\boldsymbol{n})}{b + i\boldsymbol{r}\cdot\boldsymbol{n}}\right\} e^{-i\omega_0(t-\boldsymbol{r}\cdot\boldsymbol{n})} \times \exp\left[-\frac{\omega(\boldsymbol{r}\times\boldsymbol{n})^2}{2(b + i\boldsymbol{r}\cdot\boldsymbol{n})}\right]. \quad (35)$$

Our notations here are the same as in Ref. [37]: ω_0 is the carrier frequency of the beam, b is half of the Rayleigh length, $\boldsymbol{\varepsilon}$ is a (complex) polarization vector, and $\boldsymbol{n}$ is a unit vector along the focal axis. The magnetic field is given by $\boldsymbol{H} = -(ic/\omega_0)\nabla \times$

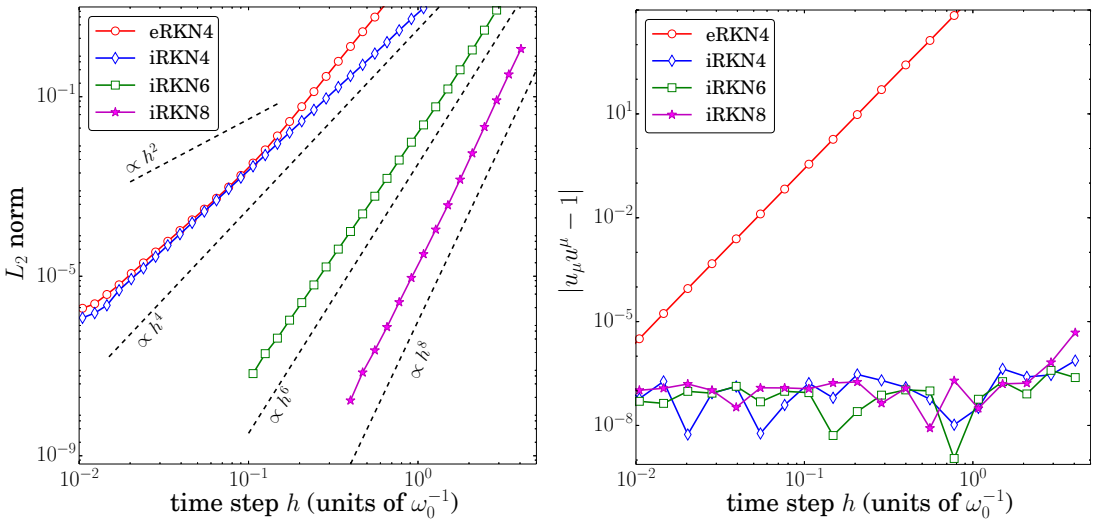


FIG. 3. (Color online) Convergence study for a particle moving in a circularly polarized plane wave for $a_0 = 10^3$. (left) Convergence in L_2 norm. (right) Error in Minkowski norm.

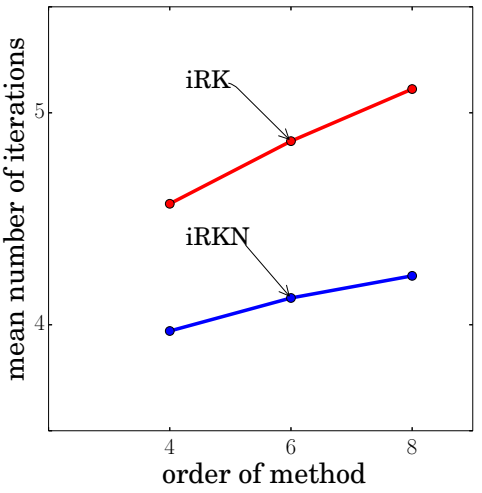


FIG. 4. (Color online) Computational complexity for a particle moving in a linearly polarized plane wave for $a_0 = 10^3$.

$\boldsymbol{E}$. Our target configuration consists of two beams with the same polarization $\boldsymbol{\varepsilon}$ but with oppositely directed vectors $\boldsymbol{n}$. In such a case near the focal axis the electric field is doubled, while the magnetic field vanishes. However, in the rest of the focal plane $\boldsymbol{n}\cdot\boldsymbol{r} = 0$, only the component $\boldsymbol{H}_\perp$ in this plane is vanishing, while its longitudinal component $\boldsymbol{H} \parallel \boldsymbol{n}$ is also doubled. Nevertheless, the component $\boldsymbol{H}_\parallel$ is of the order of $O(\Delta)$, where $\Delta = (2\omega_0 b)^{-1/2}$ is the angular aperture of the beams, which is assumed to be sufficiently small. Below, we consider a particular case of circular polarization of the beams in the sense discussed in Refs. [36,37].

The ponderomotive potential can be defined as follows [35,36]:

$$U(\boldsymbol{r}) = m\sqrt{1+\frac{e^2}{m^2\omega_0^2}|\boldsymbol{E}_{\rm tot}(\boldsymbol{r},t)|^2}, \tag{36}$$

where $\boldsymbol{E}_{\rm tot} = \boldsymbol{E}_{\boldsymbol{n}} + \boldsymbol{E}_{-\boldsymbol{n}}$ is the total electric field of counterpropagating beams. The scattering picture shown in Fig. 5 was calculated for several electrons with different impact parameters ρ and initial Lorentz factor $\gamma_0 = 10^3$, for $\boldsymbol{n}$ directed along the z axis, $\boldsymbol{\varepsilon} = \{1, i, 0\}$, and the initial momenta of all the electrons directed along the x axis.

An estimate for the particle energy (initial Lorentz factor γ_0) required to overcome the ponderomotive potential barrier without taking in account radiation reaction was formulated in [36]:

$$\gamma_0 > \sqrt{1+a_0^2}. \tag{37}$$

Even though this criterion was derived under some restrictions, e.g., for head on collision with a single beam possessing azimuthal symmetry of ponderomotive potential, and need not be recognized literally in our setup, it can still be used as a useful estimate for tuning the initial conditions close to the threshold values.

Unlike in previous sections, the problem under consideration does not admit analytical solutions, so it can only be explored by numerical simulations. The results of simulating the scattering of electrons incident in the focal plane transverse to the counterpropagating beams both with and without accounting for radiation reaction are presented in Fig. 5. If the scattering is described by the NL equation [red dashed trajectories in Fig. 5(a)], then all the electrons with $\gamma_0 = 10^3$ cross the focus without reflection. However, the radiation reaction effects lead to total reflection of the electrons from the laser focus (blue solid trajectories).

The energy spectra after scattering are depicted in Fig. 5(b). The energy distribution obtained from the NL equation is concentrated around the initial energy corresponding to $\gamma_0 = 10^3$, with a small dispersion $\Delta\gamma \sim 50$ (corresponding to deviations from the ponderomotive theory). The spectrum obtained with the LL equation is considerably shifted towards the lower energy and is peaked at $\gamma_1^{(cl)} \simeq 220$. For the

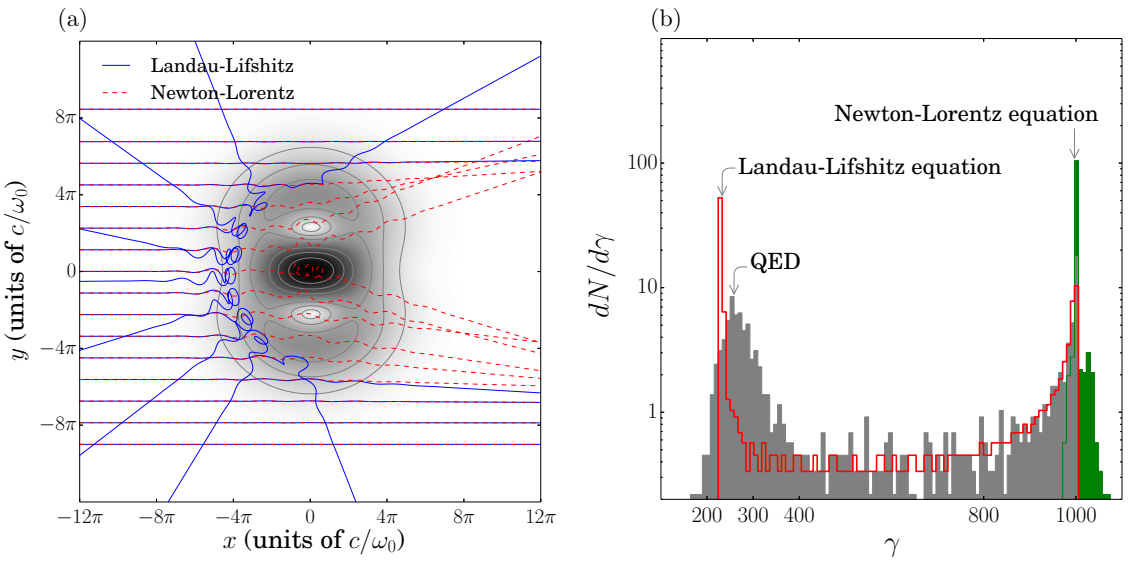


FIG. 5. (Color online) (a) Trajectories of scattered particles with radiation reaction (solid blue lines) and without radiation reaction (dashed red lines). (b) Energy spectrum of particles after they have left the focal region. The initial Lorentz factor is $\gamma_0 = 10^3$, the dimensionless field amplitude is $a_0 = 10^3$, and the focusing parameter is $\Delta = 0.1$ ($b = 31.8\lambda$).

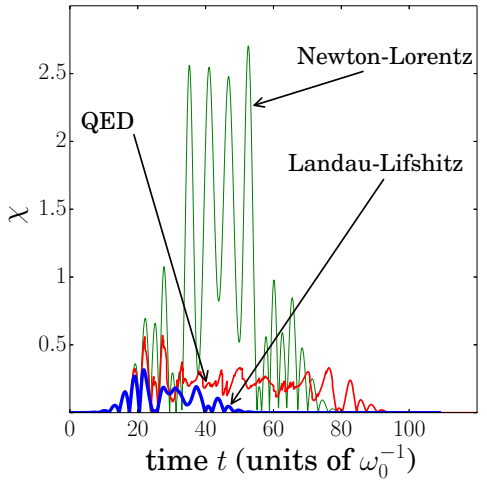


FIG. 6. (Color online) Quantum dynamical parameter χ as a function of time for a particle with an impact parameter of zero.

sake of completeness we supply this plot with the results obtained with the more fundamental method of treating radiation reaction based on quantum electrodynamics (QED). Namely, radiation reaction in this method is implemented by successive recoils due to the emission of individual photons, which are sampled using the Monte Carlo method using the probabilities of photon emission in a constant crossed field. For a discussion of the grounds of the method and further details on its implementation, see Refs. [5,14,38]. As was shown in Appendix A of Ref. [5], the QED model reproduces the leading term of the Landau-Lifshitz equation when $\chi \ll 1$.

One can see that the quantum picture results in a wider energy spectrum peaked at a slightly higher value $\gamma_1^{(q)} \simeq 250$. In order to explain why the results obtained with the LL equation and with the quantum approach deviate only slightly, although the explored value of a_0 is high, let us consider the applicability of the LL equation in more detail. As is well known, quantum and classical radiation regimes are discriminated by the quantum dynamical parameter [1]

$$\chi = \frac{e}{m^2}\sqrt{-(F^{\mu\nu}u_\nu)^2} = \frac{e}{m^2}\sqrt{(\gamma \boldsymbol{E} + \boldsymbol{u}\times\boldsymbol{H})^2 - (\boldsymbol{u}\cdot\boldsymbol{E})^2}. \tag{38}$$

Specifically, if $\chi \ll 1$, then radiation reaction is classical; otherwise, it must be treated as quantum. The parameter χ is depicted in Fig. 6 as a function of time for a particle with an impact parameter of zero for three different models (without radiation reaction, the LL equation, and a quantum description of radiation). As is clear from Fig. 6, although the naive model based on the NL equation predicts that $\chi \gtrsim 1$, because of radiation reaction, the particle strongly decelerates on its way towards the focus, and therefore χ remains small. Hence one can expect that for our values of parameters the LL equation provides qualitatively acceptable results. The notable broadening in the spectra in the QED model versus the classical RR model in Fig. 5 (right panel) comes from the stochastic nature of quantum radiation, which shows up even if the parameter χ remains moderately small. This regime can be described in momentum space by the Fokker-Planck equation (see, e.g., [13]).

In Fig. 7(a), the scattering angle θ is shown as a function of the impact parameter. The studies are performed for $N_p = 1024$ particles distributed initially uniformly at $x = -18\pi$ with the initial momenta directed along the y axis and $\gamma_0 = 10^3$. The impact parameter corresponds to the initial coordinate $-12\pi < y < 12\pi$ of the particles. The parametric studies presented in this section are performed using the eRKN4 and iRKNp ($p = 4,6,8$) methods with a time step value $h = 10^{-4}\omega_0^{-1}$. To estimate the relative accuracy in a scattering angle θ we employed the iRKN8 method. The relative numerical

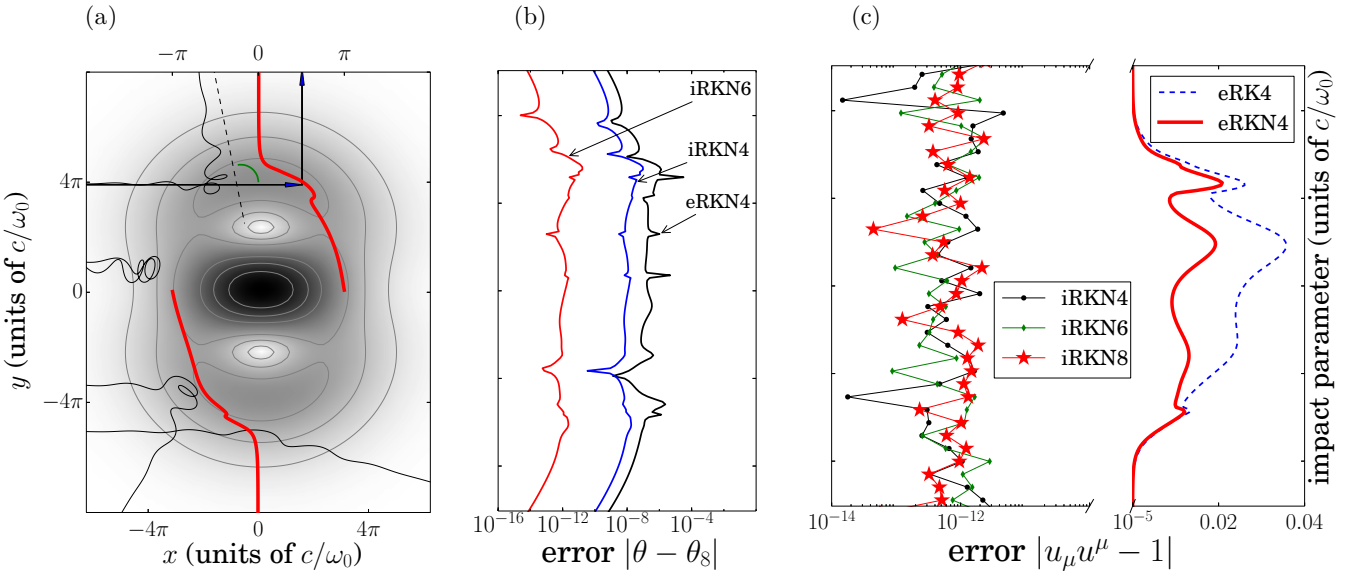


FIG. 7. (Color online) (a) Dependence of the scattering angle on the impact parameter; (b) and (c) estimations of accuracy for numerical simulation of the scattering problem using the relative scattering angle error and the Minkowski norm (parameters are the same as in Fig. 5). The simulation is performed with the time step value $h = 10^{-4}\omega_0^{-1}$.

errors $|\theta_{4,6} - \theta_8|$ in the scattering angle θ are shown in Fig. 7(b). The errors in the scattering angle have a pronounced dependency on the impact parameter. Thus the accuracy of the method is sensitive to the local gradients of the field. The error in the Minkowski norm for all the implicit methods under consideration is not sensitive to the impact parameter [see Fig. 7(c)], but the error of the explicit eRK4 and eRKN4 methods shows a pronounced dependence on the impact parameter. However, the eRKN4 method performs slightly better than eRK4 in preservation of the on-shell condition.

VII. SUMMARY AND CONCLUSIONS

In this paper we studied the numerical methods for an accurate integration of the Landau-Lifshitz equation. The Landau-Lifshitz equation is proven to be a robust and computationally cheap model for including the radiation reaction effects in particle dynamics. The result of our study showed the advantage of the Runge-Kutta-Nyström methods applied to the Landau-Lifshitz equation over the Runge-Kutta methods of the same order. The Runge-Kutta-Nyström methods are specially designed to solve the second order ODEs with fewer arithmetical operations than required for the equivalent system of the first order ODEs. One important observation is that the Minkowski norm is not always conserved in the course of the numerical simulation for both the Landau-Lifshitz and Newton-Lorentz equations. This problem is particularly strong for explicit methods, but the implicit methods keep the Minkowski norm bounded.

We derived the fourth, sixth, and eighth order Runge-Kutta-Nyström methods based on the Gauss-Legendre collocation points. These methods demonstrate excellent numerical accuracy and conservation properties. At first glance their numerical complexity is higher since they involve iterations. But, in fact, implicit methods outperform the explicit ones since they allow larger integration steps. Further speedup of the implicit RKN method can be gained if a more sophisticated iteration process is applied.

The accuracy of all the methods under consideration was thoroughly tested against two analytical solutions of the Landau-Lifshitz equation. The first benchmark case was the solution of the Landau-Lifshitz equation for a charged particle rotating in a constant magnetic field. Radiation losses in this case correspond to the well known synchrotron radiation. The second analytical solution considered was particle motion in a plane wave. We focused on a circularly polarized plane wave because in that case the phase can be explicitly expressed in terms of the proper time.

Finally, we presented the results of successful application of the RKN methods to the simulation of the scattering of high-energy electrons on focused laser beams. Our simulation indicates that taking into account radiation reaction results in strong deviation of the scattering patterns from those predicted by the ponderomotive scattering theory. We also carried out quasiclassical modeling of radiation reaction due to the emission of hard photons. The obtained results are qualitatively similar to those discovered with the Landau-Lifshitz equation. The results of the scattering test reveal a strong correlation between the numerical accuracy and the impact parameter of a scattering particle. The local loss of accuracy happens due to strong gradients of the external field. To restore accuracy an adaptive time step control can be applied. This problem has to be tackled in future studies.

We believe that our results for a Landau-Lifshitz equation solver will be useful in routine calculations for planning future laser-matter interaction experiments at high laser intensities.

ACKNOWLEDGMENTS

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APPENDIX: HIGHER ORDER COLLOCATION RKN METHOD

In the case with $s = 3$ the Lagrange basis of polynomials (15) takes the form

$$L_1 = \frac{\xi - c_2}{c_1 - c_2}\frac{\xi - c_3}{c_1 - c_3}, \quad L_2 = \frac{\xi - c_1}{c_2 - c_1}\frac{\xi - c_3}{c_2 - c_3},$$
$$L_3 = \frac{\xi - c_1}{c_3 - c_1}\frac{\xi - c_2}{c_3 - c_2}. \tag{A1}$$

Evaluation of the coefficients A_{ij}, B_{ij}, a_i, and b_i requires just elementary integration. The PYTHON script for evaluation of the iRKN6 method can be obtained from the authors upon request. The advantage of symbolic calculation is the analytical form of the coefficients formulated in terms of the collocation points c_1, c_2, c_3. For example, A_{11} and B_{11} are given by

$$A_{11} = \frac{c_1\left[2c_1^2 - 3c_1(c_2 + c_3) + 6c_2c_3\right]}{6(c_1 - c_2)(c_1 - c_3)},$$
$$B_{11} = \frac{c_1^2\left[c_1^2 - 2c_1(c_2 + c_3) + 6c_2c_3\right]}{12(c_1 - c_2)(c_1 - c_3)}. \tag{A2}$$

TABLE IV. Gauss-Legendre RKN method of order 6.

$\frac{1}{2} - \frac{\sqrt{15}}{10}$	$\frac{5}{36}$	$\frac{2}{9} - \frac{\sqrt{15}}{15}$	$\frac{5}{36} - \frac{\sqrt{15}}{30}$	$\frac{1}{120}$	$\frac{1}{12} - \frac{\sqrt{15}}{45}$	$\frac{13}{12} - \frac{\sqrt{15}}{36}$
$\frac{1}{2}$	$\frac{5}{36} + \frac{\sqrt{15}}{24}$	$\frac{2}{9}$	$\frac{5}{36} - \frac{\sqrt{15}}{24}$	$\frac{5}{96} + \frac{\sqrt{15}}{72}$	$\frac{1}{48}$	$\frac{5}{96} - \frac{\sqrt{15}}{72}$
$\frac{1}{2} + \frac{\sqrt{15}}{10}$	$\frac{5}{36} + \frac{\sqrt{15}}{30}$	$\frac{2}{9} + \frac{\sqrt{15}}{15}$	$\frac{5}{36}$	$\frac{15}{36} + \frac{\sqrt{13}}{120}$	$\frac{1}{12} + \frac{\sqrt{15}}{45}$	$\frac{1}{120}$
	$\frac{5}{18}$	$\frac{4}{9}$	$\frac{5}{18}$	$\frac{5}{36} + \frac{\sqrt{15}}{36}$	$\frac{2}{9}$	$\frac{5}{36} - \frac{\sqrt{15}}{36}$

In order to obtain the method of order 6, the coefficients must be evaluated at the Gaussian points $c_1 = 1/2 - \sqrt{15}/10$, $c_2 = 1/2$, $c_3 = 1/2 + \sqrt{15}/10$. The resulting coefficients for the iRKN6 method are summarized in Table IV.

[1] A. I. Nikishov and V. I. Ritus, Sov. J. Exp. Theor. Phys. **25**, 1135 (1967).

[2] V. N. Baier and V. Katkov, Sov. Phys. JETP **28**, 807 (1969).

[3] A. Di Piazza, K. Z. Hatsagortsyan, and C. H. Keitel, Phys. Rev. Lett. **105**, 220403 (2010).

[4] D. G. Green and C. N. Harvey, Phys. Rev. Lett. **112**, 164801 (2014).

[5] N. V. Elkina, A. M. Fedotov, I. Y. Kostyukov, M. V. Legkov, N. B. Narozhny, E. N. Nerush, and H. Ruhl, Phys. Rev. Spec. Top. Accel. Beams **14**, 054401 (2011).

[6] P. A. M. Dirac, Proc. R. Soc. London, Ser. A **167**, 148 (1938).

[7] N. P. Klepikov, Usp. Fiz. Nauk **146**, 317 (1985).

[8] F. Rohrlich, *Classical Charged Particles* (World Scientific, Singapore, 2007).

[9] L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields* (Pergamon, Oxford, 1975).

[10] A. Macchi, *A Superintense Laser-Plasma Interaction Theory Primer* (Springer, Heidelberg, 2013).

[11] G. Lehmann and K. H. Spatschek, Phys. Rev. E **85**, 056412 (2012).

[12] B. Cerutti, G. R. Werner, D. A. Uzdensky, and M. C. Begelman, Astrophys. J. **782**, 104 (2014).

[13] N. Neitz and A. Di Piazza, Phys. Rev. Lett. **111**, 054802 (2013).

[14] I. V. Sokolov, N. M. Naumova, and J. A. Nees, Phys. Plasmas **18**, 093109 (2011).

[15] A. Zhidkov, S. Masuda, S. S. Bulanov, T. Hosokai, J. Koga, and R. Kodama (2013), arXiv:1308.1608.

[16] T. Schlegel and V. T. Tikhonchuk, New J. Phys. **14**, 073034 (2012).

[17] M. Tamburini, F. Pegoraro, A. Di Piazza, C. H. Keitel, and A. Macchi, New J. Phys. **12**, 123005 (2010).

[18] R. Capdessus, E. d'Humières, and V. T. Tikhonchuk, Phys. Rev. E **86**, 036401 (2012).

[19] A. Di Piazza, C. Müller, K. Z. Hatsagortsyan, and C. H. Keitel, Rev. Mod. Phys. **84**, 1177 (2012).

[20] C. K. Birdsall and A. B. Langdon, *Plasma Simulation Via Computer Simulation*, Series in Plasma Physics (Taylor and Francis, New York, 2005).

[21] L. Verlet, Phys. Rev. **159**, 98 (1967).

[22] J.-L. Vay, Phys. Plasmas **15**, 056701 (2008).

[23] H. Kawaguchi, in *Computing Anticipatory Systems: CASYS 2001—Fifth International Conference*, AIP Conf. Proc. No. **627** (AIP, New York, 2002), 244.

[24] C. Harvey, T. Heinzl, N. Iji, and K. Langfeld, Phys. Rev. D **83**, 076013 (2011).

[25] C. Moler and C. van Loan, SIAM Rev. **20**, 801 (1978).

[26] E. J. Nyström, Acta Soc. Sci. Fenn. **50**, 1 (1925).

[27] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, Graduate Texts in Mathematics Vol. 52 (Springer, Berlin, 2002).

[28] J. Butcher, *Numerical Method for Ordinary Differential Equations* (Wiley, Chichester, UK, 2008).

[29] H. Brunner, Comput. Math. Appl. **14**, 549 (1987).

[30] The time step index n of the auxiliary stage quantities K_i^μ, L_i^μ is assumed but is omitted for the sake of simplicity.

[31] E. Fehlberg, Computing **14**, 371 (1975).

[32] E. Hairer and G. Wanner, *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems* (Springer, Berlin, 1996).

[33] A. Di Piazza, Lett. Math. Phys. **83**, 305 (2008).

[34] Y. Hadad, L. Labun, J. Rafelski, N. Elkina, C. Klier, and H. Ruhl, Phys. Rev. D **82**, 096012 (2010).

[35] D. Bauer, P. Mulser, and W. H. Steeb, Phys. Rev. Lett. **75**, 4622 (1995).

[36] N. B. Narozhny and M. S. Fofanov, Sov. J. Exp. Theor. Phys. **90**, 753 (2000).

[37] A. M. Fedotov, Laser Phys. **19**, 214 (2009).

[38] M. V. Legkov, A. M. Fedotov, N. Elkina, and H. Ruhl, Proc. SPIE **7994**, 799423 (2010).